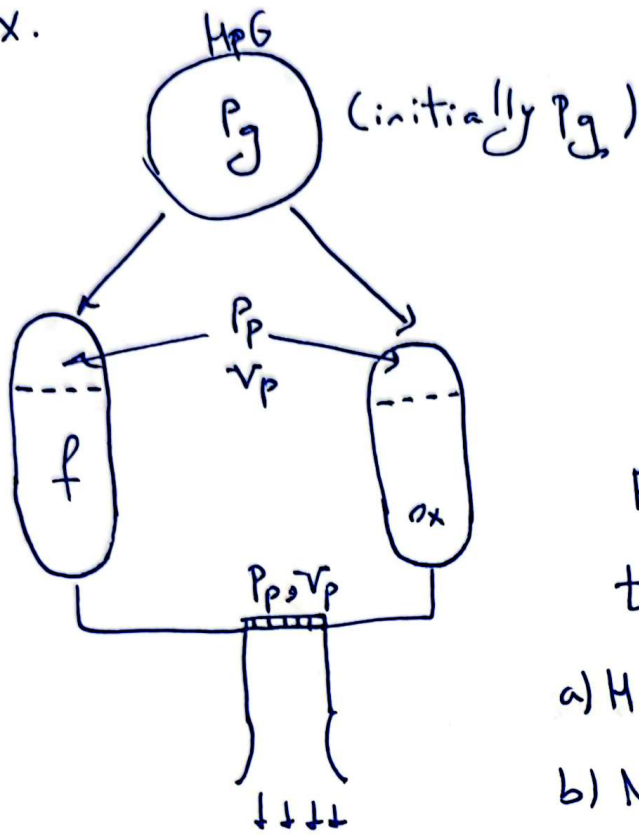


Ex.



(Sec. 4)

The question gives this data:

$$P_g = 40 \text{ MPa}$$

$$P_p = 10 \text{ MPa}$$

$$V_p = 0.28 \text{ m}^3$$

Find the minimum volume of gas tank with:

a) Helium ($\gamma = 1.67$, $\bar{M} = 4$) in Hp6

b) Nitrogen ($\gamma = 1.29$, $\bar{M} = 28$) " "

$T_g = 300 \text{ K}$, Find m_g as well

Solution: We have the data given above, first compute R

$$R = \frac{\bar{R}}{\bar{M}} = \frac{8314.3}{4} = 2078.6 \frac{\text{J}}{\text{kgK}}$$

$$V_g = \frac{\gamma P_p V_p}{P_g - P_p} = \frac{1.67 \times 10 \times 0.28}{(40 - 10)} = 0.156 \text{ m}^3$$

$$\rho_g = \frac{P_g}{R_{He} T_g} = \frac{40 \times 10^6}{2078.6 \times 300} = 64.15 \frac{\text{kg}}{\text{m}^3}$$

$$\rho_{He \text{ at S.L}} = 0.162 \frac{\text{kg}}{\text{m}^3} \text{ (compare with above!)}$$

$$m_g = \rho_g V_g = 64.15 \times 0.156 = 10 \text{ kg}$$

The required mass of He

(P-1)

As a class work, do above for N_2 :

$$R_{N_2} = 296.9 \frac{J}{kg \cdot K}$$

$$V_{g_{N_2}} = 0.120 \text{ m}^3 \quad (\text{less volume than He})$$

$$\rho_{g_{N_2}} = 449 \frac{kg}{m^3}$$

$$m_{g_{N_2}} = 53.88 \text{ kg} \quad \text{More mass!}$$

② Pump Feed System:

Objective: A formula for Pump power

useful Eqⁿ: $T ds = dh - \frac{dp}{\rho} \quad (A)$

Turbo-pump:

1st law for the pumps:

$$\dot{m}_p h_{o1} - \dot{w}_{\text{pump}} = \dot{m}_p h_{o2}$$

$$\text{or } (h_{o2} - h_{o1}) = \frac{-\dot{w}_p}{\dot{m}_p} = w_p^{\text{wp}}$$

$$\text{or } \boxed{\Delta h_o = -w_p} \quad (B)$$

An ideal pump is isentropic - Apply (A) with stagnation

condition: $\cancel{T_o} ds_o = dh_o - \frac{dp_o}{\rho_o} \Rightarrow dh_o = \frac{dp_o}{\rho_o} = \frac{dp_o}{\rho}$

$$\Rightarrow \boxed{\Delta h_{o_s} = \frac{\Delta p_o}{\rho}} \quad (C)$$

$\swarrow$
 isentropic

$\rho = \text{etc in liquids!}$

(P-2)

$$\dot{m} = (\dot{m}_{ker} + \dot{m}_{Lox}) = \dot{m}_{ker} + r \dot{m}_{ker} = \dot{m}_{ker} (1+r)$$

$$92 = \dot{m}_{ker} (1+2.5) \Rightarrow \dot{m}_{ker} = \frac{92}{3.5} = 26.3 \frac{\text{kg}}{\text{s}}$$

$$\dot{m}_{Lox} = r \dot{m}_{ker} = 2.5 \times 26.3 = 65.7 \frac{\text{kg}}{\text{s}}$$

$$|P_p| = |P_p|_{ker} + |P_p|_{Lox} = \left[\frac{\dot{m} \Delta P_o}{\eta_p \rho} \right]_{ker} + \left[\frac{\dot{m} \Delta P_o}{\eta_p \rho} \right]_{Lox}$$

$$\rho = SG \times \rho_w \Rightarrow \begin{cases} \rho_{ker} = 0.8 \times 1000 = 800 \frac{\text{kg}}{\text{m}^3} \\ \rho_{Lox} = 1.14 \times 1000 = 1140 \end{cases}$$

$$|P_p| = \left[\frac{26.3 \times 3.8 \times 10^6}{0.65 \times 800} \right] + \left[\frac{65.7 \times 4.6 \times 10^6}{0.65 \times 1140} \right]$$

$$(1.92 + 4.08) \times 10^5 = \boxed{0.6 \text{ MW}}$$